Assignment 2

1. Question 1 — BridgeAndFerry

1.1 Problem Formulation

BridgeAndFerry is a 7×11 episodic grid-world. The agent starts at S=(3,0) and aims to reach G=(3,10). The central bridge occupies row 3, columns 3–7 and is exposed to an independent wind event whenever the agent leaves a bridge cell. A wind event displaces the agent vertically into the adjacent chasm and terminates the episode. The ferry at F=(0,5) terminates the episode with a stochastic reward, while a longer southern detour can reach the goal without crossing the bridge.

The implementation uses 77 discrete states (7×11) and four actions: Up, Down, Left and Right. The step cost is −1 on every transition, including terminal transitions. Reaching the goal gives +50 before the step cost (+49 net), falling into the chasm gives −100 before the step cost (−101 net), and reaching the ferry gives $N(45,\sigma^2)-1$. $\gamma=1$ is also fixed value.

1.2 Implementation and Experimental Setup

Parameter	Value
Algorithms	SARSA, Expected SARSA, Q-Learning
Behaviour policy	ε-greedy, ε=0.1
Step size	α=0.1
Discount	γ=1.0
Main training	20,000 episodes / algorithm
Threshold + variance sweeps	8,000 episodes / configuration
Seeds	10 independent seeds
Greedy evaluation	500 episodes / seed
Episode cap	500 steps
Initial Q-values	$Q_0=50$

1.3 Main Experiment: p_wind=0, σ=0

At zero wind, the expected route returns are +40 for the bridge, +37 for the ferry and +34 for the southern detour. All three algorithms were trained under the same schedule and the frozen greedy policy was evaluated over 500 episodes per seed.

Algorithm	Mean greedy return	Non-terminating eval episodes	Fully non-terminating seeds
SARSA	37.00 ± 0.00	500 / 5000	1 / 10
Expected SARSA	37.00 ± 0.00	0 / 5000	0 / 10
Q-Learning	40.00 ± 0.00	0 / 5000	0 / 10

Q-Learning develops a strong bridge-following visitation pattern, while SARSA and Expected SARSA show much heavier visitation of the ferry region. The Q-Learning greedy policy reaches the +40 bridge return in the final evaluation. SARSA has one seed whose greedy policy does not terminate within the 500-step cap; that seed is reported separately.

1.4 State-Visitation Analysis

The heat maps count visited cells during training, including the start and terminal cells, and report average visitation count per episode across the ten seeds. Q-Learning concentrates visitation along row 3,

consistent with learning the bridge route. SARSA and Expected SARSA show stronger visitation around the ferry route and substantially less traversal of the bridge corridor.

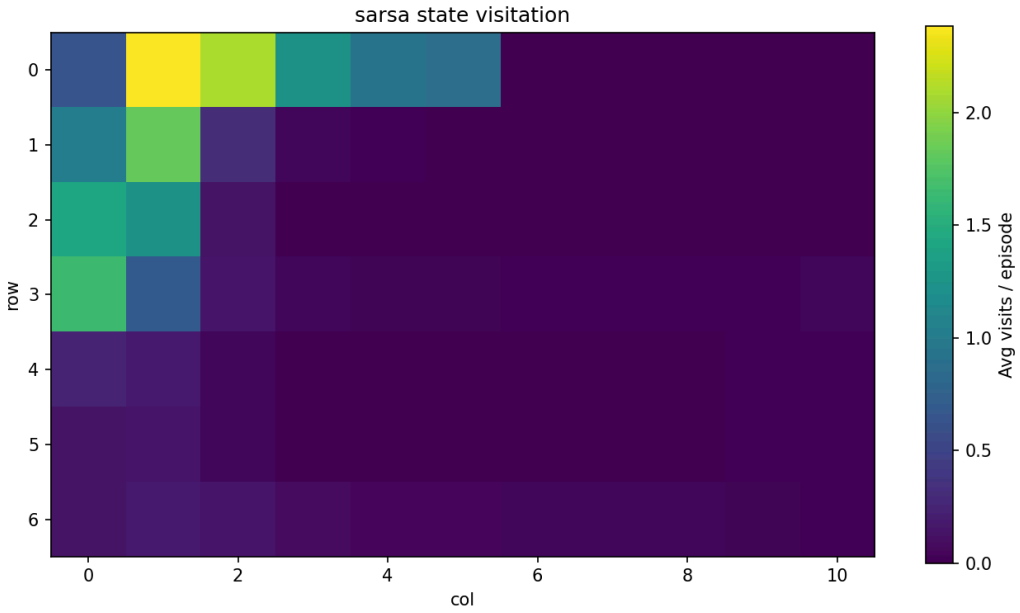


Figure 2. SARSA state-visitation heat map.

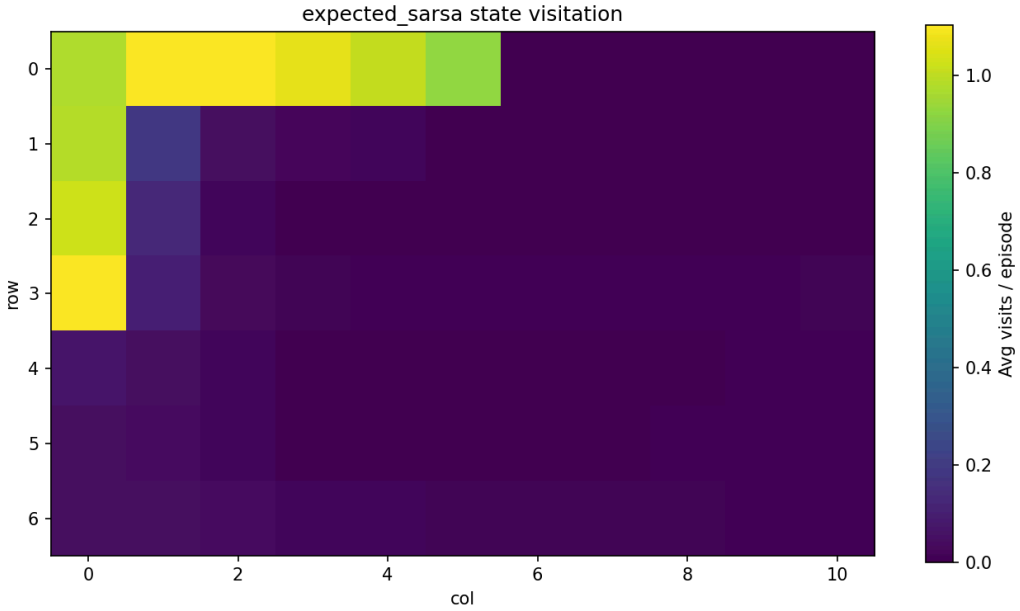


Figure 3. Expected SARSA state-visitation heat map.

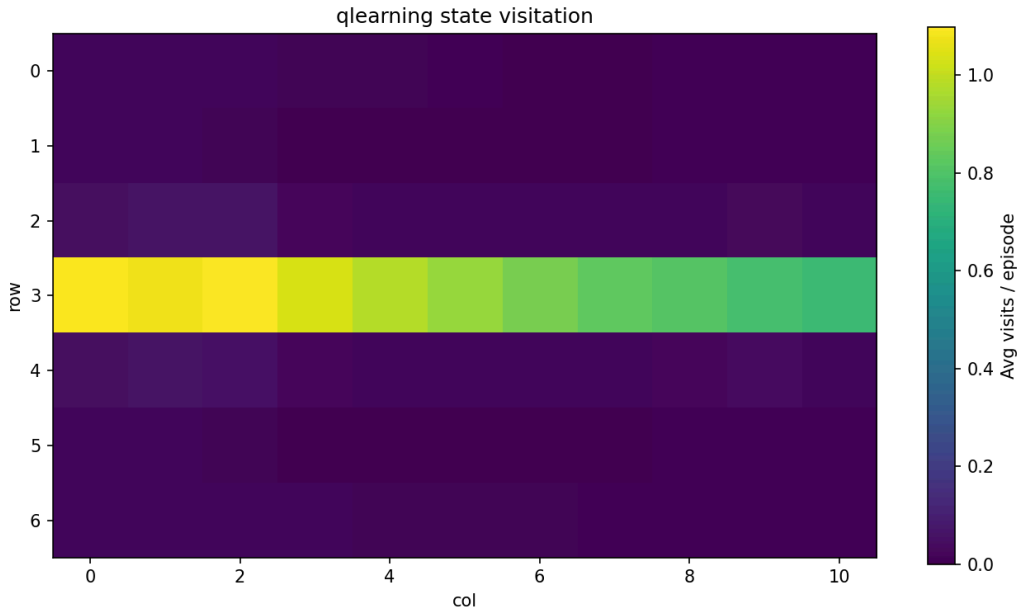


Figure 4. Q-Learning state-visitation heat map.

1.5 Analytical Bridge Threshold

The bridge route succeeds with probability (1−p)^5 because five bridge-cell wind checks must all be survived. Full success has return +40. If the first wind event occurs at the kth bridge check, the agent has already taken 3+(k−1) successful actions and the fatal transition contributes −101.

E[G_bridge] = 40(1−p)^5 + Σ(k=1..5) p(1−p)^(k−1)[−(3+k−1)−101].

The ferry has expected return +37, independent of σ because its payout mean remains 45. Solving E[G_bridge]=37 gives p*=0.004144 (approximately 0.414%). The expected bridge returns at p={0.002,0.004,0.008} are approximately {38.546, 37.103, 34.253}. Thus the theoretical switch is very close to p=0.004.

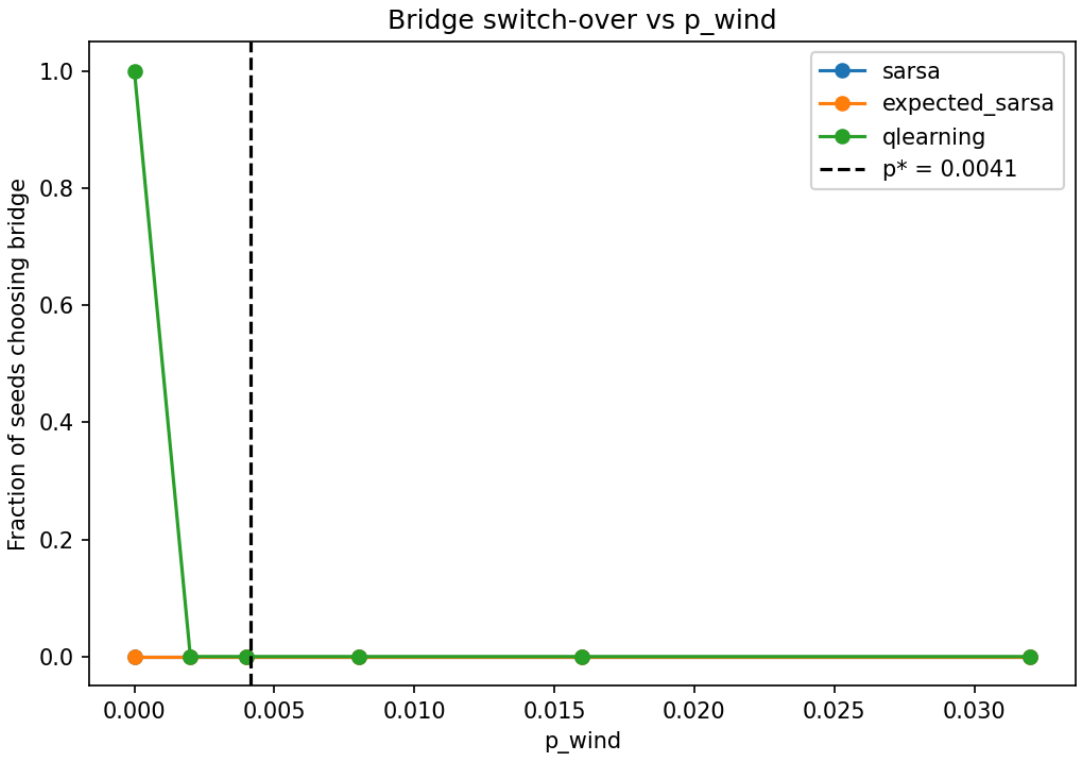


Figure 5. Empirical bridge-selection fraction with the analytical threshold marked.

1.6 Empirical Switch-Over Experiment

The sweep uses p_wind={0,0.002,0.004,0.008,0.016,0.032}, ten training seeds per configuration, and 8,000 training episodes.

p_wind	SARSA	Expected SARSA	Q-Learning
0	0/10	0/10	10/10
0.002	0/10	0/10	0/10
0.004	0/10	0/10	0/10
0.008	0/10	0/10	0/10
0.016	0/10	0/10	0/10
0.032	0/10	0/10	0/10

The empirical sweep does not exhibit a switch-over for SARSA or Expected SARSA within the tested grid. Q-Learning switches between p=0 and p=0.002 in this run, which is not close to the analytical threshold p*=0.004144. This should therefore be interpreted as a limitation of the finite learning/exploration setup rather than evidence that the theoretical threshold is different.

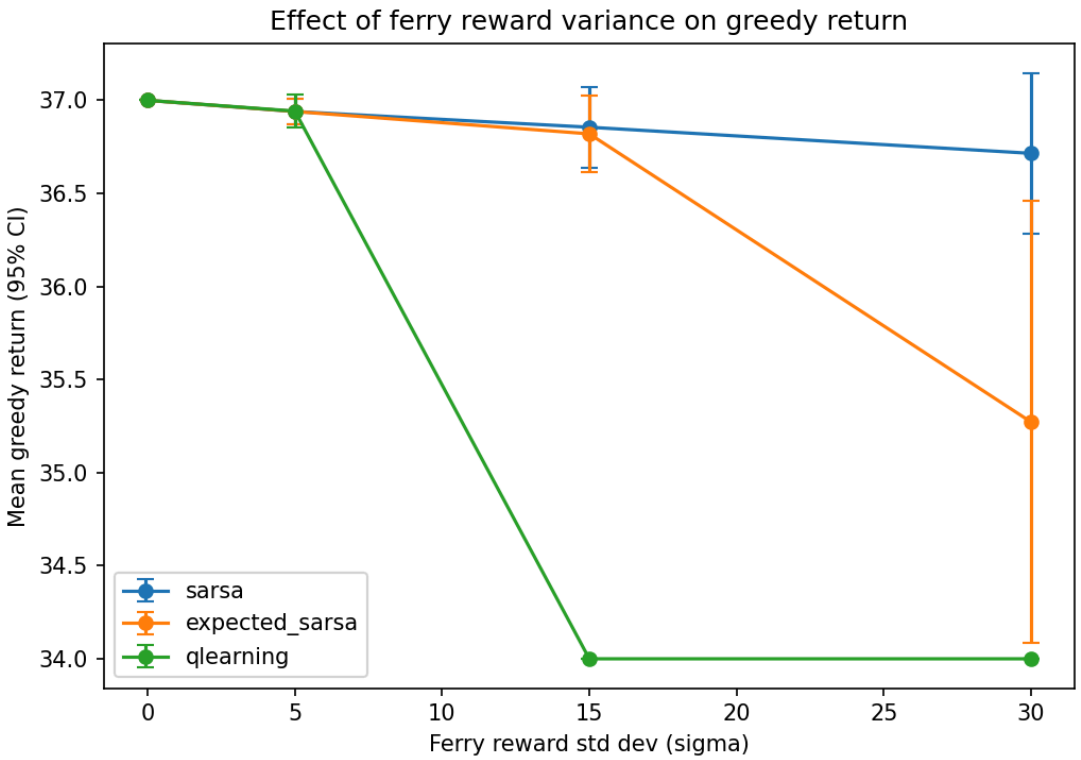


Figure 6. Mean greedy return versus ferry reward variance, with 95% CI.

1.7 Effect of Ferry Reward Variance

At p_wind=0.02, the ferry's mean return remains +37 for every σ, while its variance increases with σ. The experiment uses σ={0,5,15,30}, ten independent training seeds, and 500 greedy evaluation episodes per seed.

Algorithm	σ	Ferry fraction	Mean greedy return	Non-term. seeds
SARSA	0	0.70	37.00 ±0.00	3
SARSA	5	1.00	36.94 ±0.07	0
SARSA	15	0.90	36.86 ±0.22	1
SARSA	30	0.90	36.72 ±0.43	1
Expected SARSA	0	1.00	37.00 ±0.00	0
Expected SARSA	5	1.00	36.94 ±0.07	0
Expected SARSA	15	1.00	36.82 ±0.21	0
Expected SARSA	30	0.30	35.27 ±1.19	3
Q-Learning	0	1.00	37.00 ±0.00	0
Q-Learning	5	0.80	36.94 ±0.09	2
Q-Learning	15	0.00	34.00 ±0.00	0
Q-Learning	30	0.00	34.00 ±0.00	0

The clearest reduction in ferry selection occurs for Q-Learning: its ferry fraction falls from 1.00 at σ=0 to 0 at σ=15 and remains 0 at σ=30. Expected SARSA remains ferry-dominant through σ=15 and falls to 0.30 at σ=30. SARSA remains ferry-dominant across the tested variances, with several non-terminating seeds.

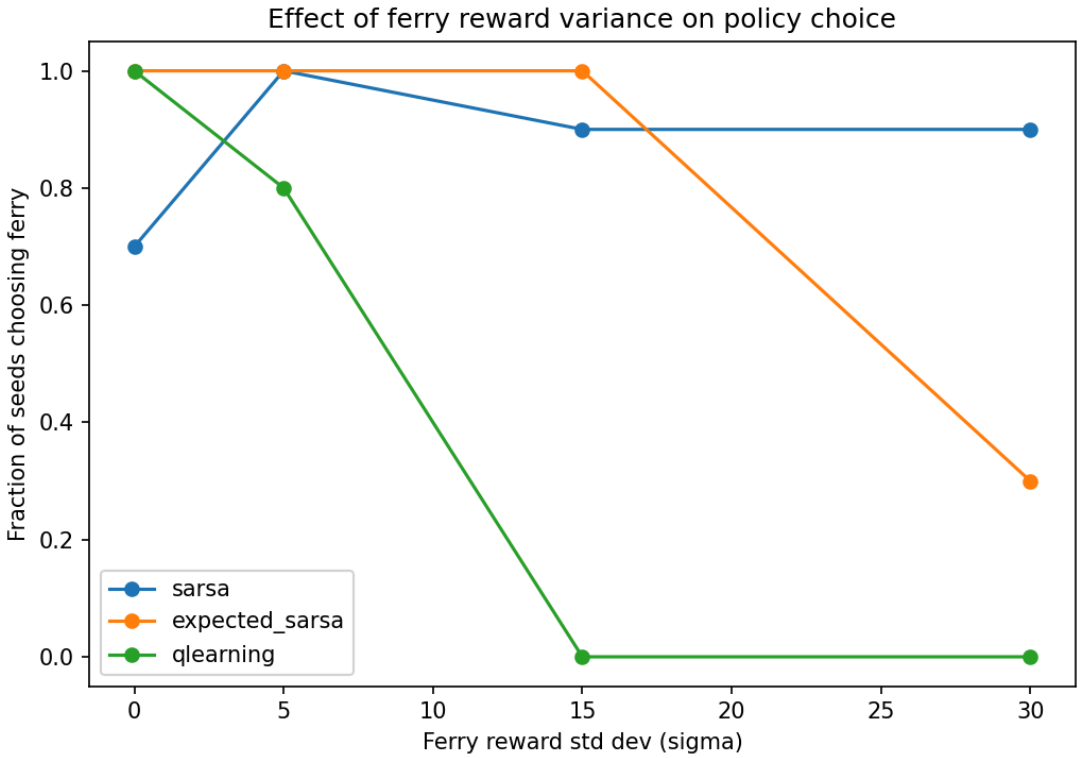


Figure 7. Fraction of training seeds whose final greedy policy chooses the ferry.

1.8 Interpretation and Conclusion

At p_wind=0, Q-Learning is the only algorithm in the current run that consistently learns the narrow bridge route. Its visitation pattern reflects repeated traversal of row 3, whereas SARSA and Expected SARSA predominantly learn the ferry policy. The analytical threshold provides an environment-level reference independent of the learning algorithm. The mismatch between p*=0.004144 and the empirical switch-over indicates that the current finite training procedure does not reliably resolve the small value difference near the threshold.

The variance results show different empirical responses to noisy ferry rewards. These observations describe behaviour under the specified finite training budget and seeds rather than a universal ranking of the algorithms.

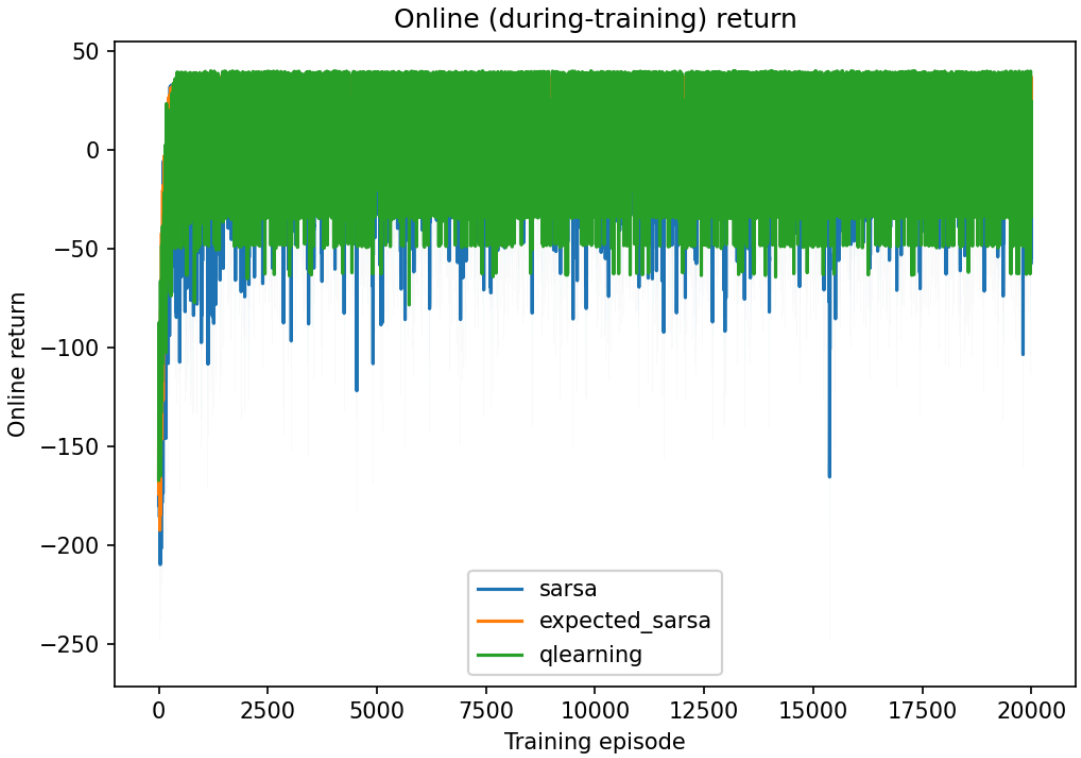
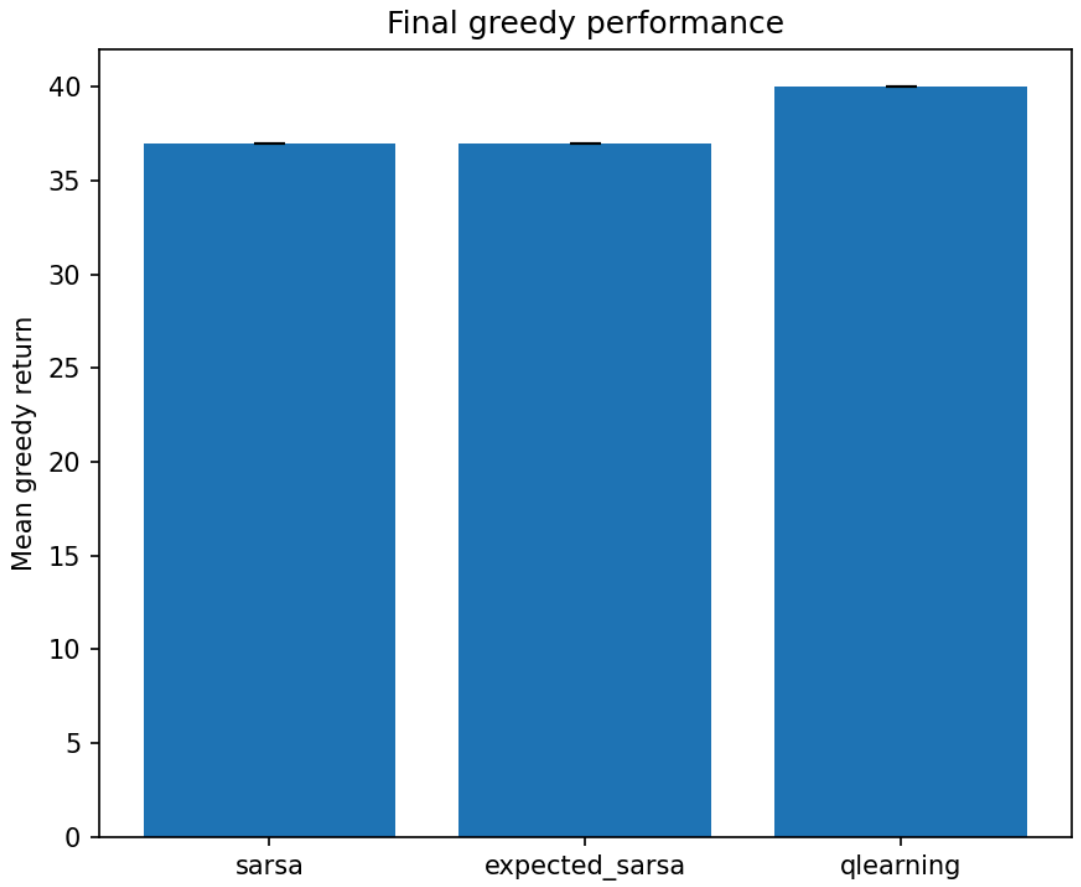


Figure 1. Online return during training; mean across 10 seeds with ±1 standard error.



2. Question 2 — SupplyRun

2.1 Problem Formulation

SupplyRun is an 8×8 deterministic grid-world. The agent starts at S=(7,0), must first reach the depot D=(0,7) to collect cargo, and must then reach the drop-off X=(7,7). The refuelling cells are R1=(1,2) and R2=(4,4), each restoring fuel to capacity when visited.

The implementation state is (row, column, has_cargo, fuel), and the action space contains Up, Down, Left and Right. Each action consumes one unit of fuel. Moving into a wall leaves the agent in place but still consumes fuel. Reaching the depot sets has_cargo=1. Reaching X with cargo gives +100 and terminates; exhausting fuel gives −20 and terminates; all other transitions give 0 reward.

The environment can otherwise continue indefinitely because an agent can shuttle between refuelling cells. Therefore episodes are truncated at 200 steps. A truncated episode is not a terminated episode and should retain a bootstrap value rather than assigning zero terminal value.

2.2 Learning Methods

The implementation uses tabular n-step SARSA for $n \in \{1,2,4,8,16,24,32\}$. A one-step Q-Learning agent is trained as the off-policy baseline. Both algorithms are implemented directly in nstep_agents.py.

n-step SARSA stores state, action and reward sequences and applies multi-step returns after the appropriate delay. Q-Learning uses the one-step target $r + \gamma \max_a Q(s',a)$ on non-terminal transitions.

2.3 Experimental Configuration

Parameter	Value used in supplied code
Training episodes per configuration	8,000
Independent seeds	10 (0–9)
Discount factor γ	0.99
Step size α	0.1
Behaviour ε	0.1
Episode truncation	200 steps
n values	1, 2, 4, 8, 16, 24, 32
Q initialization	Optimistic: Q=30.0
Final evaluation in supplied code	50 greedy episodes/seed — provisional

And also fixed 8,000-episode training budget, 10 seeds and γ=0.99

2.4 Truncation Handling

The environment exposes separate terminated and truncated flags. The learner contains a dedicated truncation branch intended to bootstrap rather than assigning zero value at the 200-step boundary. This distinction is essential because treating truncation as termination changes the learning problem and can produce incorrect learning curves.

2.5 Preliminary n-Step Results

The numerical results below are from the supplied implementation and were evaluated over 50 greedy episodes per seed. They are therefore preliminary; the final evaluation should use at least 500 greedy episodes per seed.

n	Mean greedy return	Std. dev.	Seed-level returns
1	100.00	0.00	100 × 10
2	100.00	0.00	100 × 10

4	100.00	0.00	100 × 10
8	40.00	48.99	100,100,0,0,100,0,0,0,0,0
16	0.00	0.00	0 × 10
24	0.00	0.00	0 × 10
32	0.00	0.00	0 × 10
Q-Learning	100.00	0.00	100 × 10

The preliminary run therefore does not establish a unique optimal n: n=1, 2 and 4 tie at a mean final return of 100. n=8 succeeds on four of ten seeds, while n=16, 24 and 32 fail across all ten seeds. Q-Learning reaches 100 on all ten seeds in this preliminary evaluation.

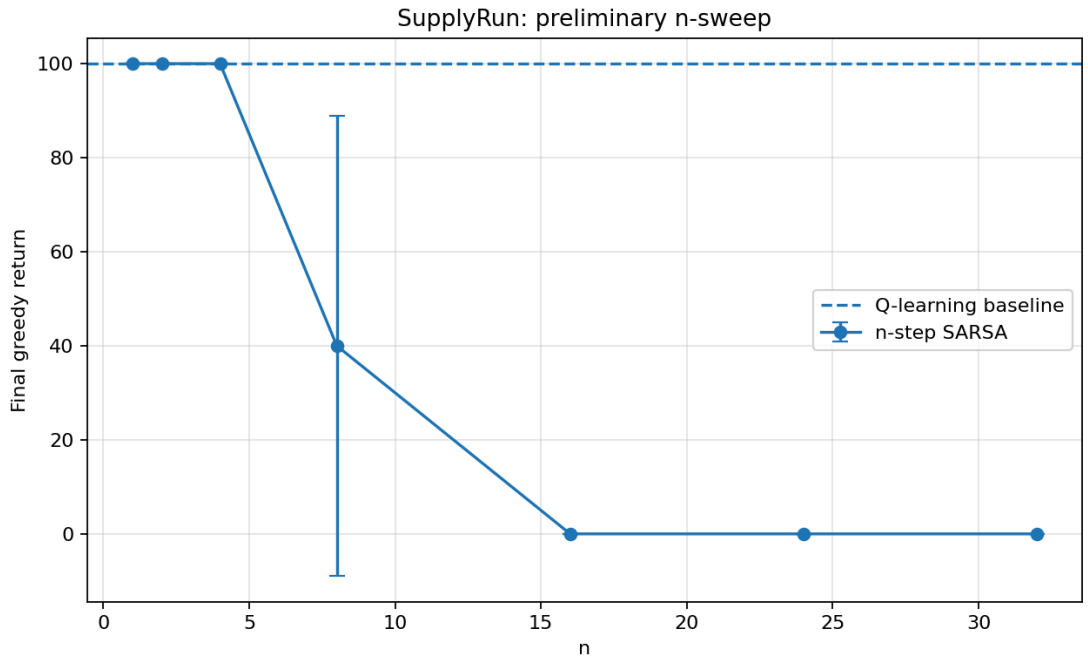


Figure 9. Preliminary final-return sweep for n-step SARSA and Q-Learning.

2.6 Interpretation

SupplyRun contains sparse delayed rewards: successful delivery requires reaching the depot and then the drop-off, while fuel exhaustion produces −20. Increasing n allows reward information to propagate over a longer temporal window, but also increases the temporal span and potential variance of the update. The preliminary results show that small n values perform well while larger n values degrade sharply.

The present data do not justify claiming a single optimal n because three tested values tie.

2.7 Overall Question 2 Conclusion

The supplied implementation constructs the required SupplyRun environment, implements n-step SARSA and one-step Q-Learning, uses the required 8,000-episode budget and 10 seeds, and evaluates a spectrum of n values. The preliminary results indicate successful learning for n=1,2,4 and the Q-Learning baseline, with deterioration for larger n. The final report should replace the provisional 50-episode evaluation with a compliant 500-episode evaluation and include the final learning-curve and n-sweep figures.